

Technology Stack Template

Laptop Request Catalog Item

Technology Stack: Laptop Request Catalog Item

Technical Architecture

The solution architecture is centered on the ServiceNow platform, leveraging its native capabilities to deliver a dynamic, governed Service Catalog item.

Component	S.No.	Description	Technology
User Interface	1.	Employee interacts with the Request form via a web browser.	ServiceNow Service Portal / Web UI
Application Logic-1 (Front-End)	2.	Logic to automatically pre-fill user details (Manager, Department) upon form load.	ServiceNow Catalog Client Script (g_form, GlideAjax)
Application Logic-2 (Front-End/Dynamic)	3.	Logic to implement dynamic field visibility, mandatory fields, and read-only status based on user selection.	ServiceNow Catalog UI Policies

Application Logic-3 (Front-End/Custom)	4.	Logic to execute the custom " Reset Form " functionality when the UI Action button is clicked.	ServiceNow Catalog Client Script (Custom Function)
Application Logic-4 (Back-End)	5.	Logic to handle the fulfillment process upon form submission (creating tasks, approvals).	ServiceNow Workflow / Flow Designer
Database	6.	Stores all request data (REQs, RITMs, Variables, and User Profile data).	ServiceNow Platform Database (CMDB, sys_user, sc_req_item tables)
Configuration Management	7.	Tool used to track all changes for deployment across Dev, Test, and Prod instances.	ServiceNow Update Set
External API	8.	(Optional) Integration for real-time inventory checks or detailed cost center validation.	ServiceNow REST API

Application Characteristics

This table defines the quality attributes of the solution, primarily based on the robust nature

of the underlying ServiceNow platform.

S.No	Characteristics	Description	Technology
1.	Security Implementations	Ensures only authenticated users can access the form and implements field-level security for sensitive data.	ServiceNow ACLs, User Roles
2.	Scalable Architecture	The solution is built on ServiceNow's SaaS platform, ensuring high performance regardless of user volume.	ServiceNow Cloud Architecture
3.	Availability	The system is highly available and accessible to all employees when the ServiceNow instance is operational.	Load-balanced ServiceNow Instances
4.	Maintainability	All custom code (Scripts) is isolated within the application scope and is fully contained within an Update Set.	Scoped Application, Clear Script Comments

5.	Performance	Front-end dynamic logic is optimized to execute rapidly using Catalog UI Policies to maintain a quick user experience.	Optimized Client Scripting
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Reference

ServiceNow Documentation: Service Catalog Development
ServiceNow Documentation: Client Script Best Practices